The Welfare Impacts of Large-Scale Land Transactions: Evidence from Ethiopia

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Abstract

Large-scale land transactions (LSLTs) have expanded rapidly across developing countries over the past decades, especially in Ethiopia. While LSLTs are often promoted as instruments for rural transformation, promising employment creation, technology transfer, and improved market integration, they also raise concerns about displacement, reduced access to natural resources, and weakened customary tenure. We combine two analyses to study the medium-run effects of LSLTs in Ethiopia. First, we leverage the plausibly exogenous timing of LSLT entry to estimate the effects on local economic and environmental outcomes. Second, we conduct an original survey of households in Gambela region—an LSLT hotbed—including detailed questions on households' interactions with LSLTs to shed light on mechanisms. Here we rely on the fact that potential investors relied on observable factors such as soil characteristics, road access, and proximity to conflict zones to identify causal effects using double-debiased machine learning. We find modest effects on crop productivity at the national level with no effect on deforestation, and significant reductions in burned area. In Gambela, we find similar effects on crop productivity and burning but also that LSLTs reduced deforestation and increased tree cover. Despite limited labor market and other direct linkages, we observe a 13.7% increase in log wage income for every 1,000 ha of LSLT within 5–10 km. We also find noisy positive effects on crop and total income at this distance, but no effects on consumption. In contrast,households within 0–5 km show no significant benefits. This is consistent with modest positive equilibrium effects that at least partially offset damages to the most exposed households.

1 Introduction

Large-scale land transactions (LSLTs) have become increasingly common in developing countries, with millions of hectares transferred from governments to private investors in recent years. These transfers have been promoted as a win-win, allowing uncultivated land to be put to productive uses while providing labor market opportunities, technology transfers and aggregate demand increases to host communities. However, such transactions have often been referred to as "land grabs" due to reports of illegally expropriating farmland, environmental degradation and deforestation (Anti, 2021). Identifying these positive and negative potential effects, which may occur simultaneously, and their incidence is therefore crucial for policy. However there is limited credible causal evidence on these channels: In particular, well-identified analyses with administrative panel data often lack the ability to directly observe how households interact with and are affected by LSLTs.

We study the potential for both of these channels in Ethiopia by combining two research designs: a national-level event-study leveraging plausibly exogenous timing in the entry of LSLTs across woredas (districts) and an ex-post analysis using original survey data of households in Gambela region, which was highly exposed to LSLTs. In particular, our survey asks farmers to detail their direct interactions with LSLTs—including land conflict and expropriation, employment, technology transfer , and other economic transactions. This also allows us to estimate heterogeneous treatment effects with regards to proximity to LSLTs, which is crucial for understanding their distributional effects. In particular, environmental exposure and direct economic engagement may decay with distance to the LSLT concession while other economic effects may be experienced more uniformly across space.

At the national level, we find the first LSLT entry into a woreda generates modest increases of +0.0269 (0.06 s.d.) in net primary productivity (NPP), a remote-sensed measure of agricultural productivity, but no effects on yield or input use in survey data. We also find no effect on forest loss, but a -0.0048 (-0.04 s.d.) decrease in EVI and a -0.0254 (-0.25 s.d.) reduction in burned area. We find similar effects using more granular definitions of treatment in Gambela: increase in NPP of +0.0759, reduction in burned area of -0.0378, as well as an increase in tree cover of +4.510.

Our study makes two major contributions. First, we contribute to a growing but methodologically limited literature on the economic impacts of LSLTs. A meta-analysis by Akalu et al. (2025) identifies 143 relevant studies, of which only 39 are quantitative; among these, 74% rely on propensity-score matching with cross-sectional data, where matching on post-treatment covariates raises identification concerns. Table 6 summarizes the key quantitative studies. Even the three DiD studies use two-way fixed effects, despite the compounding effects of early-treated units in later periods we find representing the classic case in which negative weighting bias occurs. Our study addresses this gap by exploiting the staggered rollout of Ethiopia's LSLT program across woredas, using heterogeneity-robust DiD estimators—primarily the Callaway-Sant'anna estimator (Callaway & Sant'Anna, 2021) that accommodate staggered adoption with treatment effect heterogeneity. For the Gambela survey, which is by necessity a post-treatment cross sectional analysis, we improve upon PSM by applying state of the art DML (Chernozhukov et al., 2018). Importantly, we can validate this against remote-sensed outcomes for which we do have baseline data.

Second, our original survey of 2,389 households in Gamble provides a unique look in-

side the "black box" of how households interact with LSLTs. Directly observing these reported interactions—which include limited labor market engagement and few other economic linkages—provides a lens through which to interpret our modest to moderate treatment effects, which are non-monotonic in distance to LSLTs. In particular, we interpret these as reflecting costs of direct proximity to LSLTs despite positive ambient treatment effects. We believe that we are this first to document this with a rigorous quantitative framework.

2 Background: Ethiopian Land Policy and Large-Scale Agricultural Investment

2.1 Evolution of Large-Scale Agricultural Land Investment Policy

For most Ethiopians, agricultural land is the foundation of livelihoods, contributing significantly to household welfare both directly and indirectly. Agricultural land has also been one of the defining factors of social status and, in some cases, political power throughout Ethiopian history: families with large landholdings have historically been associated with high social and political standing. The role of agricultural land has remained crucial in modern Ethiopia as it is a major source of raw materials and direct input to the growing non-agricultural sectors (manufacturing and services), although the contribution of the agricultural sector to the economy is diminishing. Hence land policies and reforms have taken center stage in Ethiopian agricultural policy, particularly in the modern era. Historically, Ethiopia has formulated a number of land policies and undertaken a number of land reforms with the objectives of bringing about suitable concurrent economic and political transformation (Dunning, 1971), often in response to public demand and changes in government (Kebede, 2008). Hence, land reform has been seen as one of the prerequisites for national growth (Dunning, 1971).

During the imperial regime's reign to 1974, the Derg military regime from 1974 to 1991, and the post-Derg regime since 1991, Ethiopia has experienced various land tenure arrangements (Bodurtha, 2011). Historically, large farm investment projects have been in operation since the imperial period in Ethiopia (Ali et al., 2019). However, the approaches, priorities, and intensity of investments vary between regimes. Although cash crop production under private investors was prioritized before the 1974 revolution, the Derg regime nationalized large-scale private commercial farms and converted them into state-produced staple foods until the Ethiopian People's Revolutionary Democratic Front (EPRDF) took power in 1991 (Ali et al., 2017; Kassie et al., 2024).

In the imperial regime, land regulations allowed for the transfer, market, and mortgage of private properties, allowing for a more equitable allocation of land among landowners in different places (Gebregziabher & Holden, 2011; Beyene Leta et al., 2021). In this regime, the country experienced two land tenure systems: the *rist* and *gult* (Teklu, 2004). The communal rist system divided and distributed land by the community according to lineage claims (Bodurtha, 2011). Gult is the non-hereditary claim to all, or a portion of the tribute owed to the king by the land's occupier (Dunning, 1971; Gebregziabher & Holden, 2011; Beyene Leta et al., 2021).

The Derg regime's Land to the Tiller land reform in February 1975 eliminated all feudal

land laws and turned land over to the public (Bodurtha, 2011; Ambaye, 2015). The February 1975 land reform declaration nationalized all rural land, replacing the imperial period's age-old land tenure with a system in which the state owns all land and gives it to farmers on a use-right basis. State farms were created from commercial large-scale modern farms. On about 800 hectares, peasant associations were formed to allot land to farmers in their authority based on family size (Kebede, 2008). The Derg nationalised rural and urban land in 1975 and redistributed land with a maximum holding of 10 hectares per farm (Holden & Yohannes, 2002). However, the regime's practices resulted in unintended consequences such as farmland fragmentation through redistribution, eviction of rural populations through forcible villagization, the inefficiency of farmer cooperatives, failure of agricultural productivity, and total poverty (Kebede, 2002; Bodurtha, 2011).

Following the fall of the Derg regime, the newly formed transitional government led by the EPRDF announced the continuation of the Derg regime's land policy through an amendment, and state ownership of land was established in the new constitution in 1995 (Deininger et al., 2003; Hagmann, 2006). Land lease limits, labor hiring on a private farm, sharecropping, and individuals holding more than 10 ha were all banned under the present land policy. Government allocation, inheritance, gift, and land leasing are all options for obtaining rural land, although it cannot be sold, swapped, or mortgaged (Bodurtha, 2011).

In Ethiopia, large-scale agricultural land investment was officially introduced during the third (1968–73) five-year plan of the imperial regime. Several investors rented land from the government and developed commercial farms until the Derg confiscated commercial farms to manage and expand state farms, which were abandoned meaningfully post-1991 when the EPRDF recognized the decisive role of private capital expansion in the development of large-scale modern farming (Ali et al., 2017; Kassie et al., 2024). Agricultural Development-Led Industrialization (ADLI) designed in 1993 prioritized enhancing the productivity of smallholder farming and expansion of large-scale private commercial farms. Private agricultural investments were intensified during the first 5-year period of the GTP (2010–2015), when the Ethiopian Agricultural Investment Land Administration Agency was established in 2013.

Large-scale land transactions in Ethiopia have been increasing, particularly in the lowland regions of the country. These areas, traditionally inhabited by pastoralist and agro-pastoralist communities, have historically experienced limited state control and governance (Moreda, 2015). Much of the land offered is classified by the state as 'unused' or 'underutilised', overlooking the spatially extensive use of land in shifting cultivation and pastoralism (Moreda, 2017). Most of the earmarked and transferred agricultural land occurs in four of the nine regional states: Benishangul-Gumuz, Gambela, Oromia, and Southern Nations Nationalities and Peoples (SNNP), with greater intensity in the Benishangul-Gumuz and Gambela regions. The government suddenly suspended the program in 2016 without transferring all land designated for large-scale agricultural investments (Hajjar et al., 2020; Moreda, 2017).

2.2 Dynamics of Large-Scale Land Investment in Ethiopia

The Ethiopian government designated over 11.5 million hectares of land for large-scale investment purposes. Between 2005 and 2015, approximately 2.47 million hectares were transferred to investors, primarily from the earmarked land. The second Growth and Transformation Plan (GTP II, 2015/16–2019/20) targeted an increase in total area of land verified for large-

scale agricultural investment from 1.1 million hectares in 2014/15 to 2.443 million hectares, with nationwide total land transfer projected to reach 4.315 million hectares by the end of 2020 (MoFED, 2015).

In the Gambela region, a peak in transfers of land occurred between 2012 and 2015 (MoFED, 2015), aligning with Ethiopia's first Growth and Transformation Plan (GTP I). This was followed by a steep decline post-2016 due to the government's abrupt suspension of the land transfer program (Hajjar et al., 2020; Moreda, 2017; Rulli & D'Odorico, 2014). In Gambela, large-scale land transfers have occurred in ten out of twelve woredas of the region. The Anuak Zone leads in both the number of investors and total land area leased, followed by the Nuer Zone and the Majang Zone. At the woreda level, Gambela Zuria Woreda holds the highest number of investors (200), followed by Itang Special Woreda (137) and Abobo Woreda (130).

3 Data

Our analysis draws on four data sources, summarized in Table 1. Summary statistics for the main outcomes and treatment measures from each source are provided in Tables 7-9.

3.1 Treatment: LSLT Parcels and Timing

Treatment is defined by the timing of LSLT entry at the woreda (district) level nationally, and at the kebele (ward) or pixel level in Gambela. We use the universe of approved investments from the Ministry of Agriculture, which provides information on the woreda, size, year of entry and name of the investor for each LSLT. This records 147 treated woredas with first treatment between 1995 and 2016. Total LSLT area across all treated woredas is 966,026 hectares, with a median of 650 ha per woreda. In Gambela, we use the shapefile with the exact geocoordinates of LSLT parcels. This covers 36 treated kebeles (treatment years 2000–2014), which we match to the Ministry of Agriculture records to obtain the year of treatment. Figures 1(a) and 1(b) maps the spatial and temporal distribution of treatment at both levels.

3.2 TODO Remote-Sensing Outcomes (GEE)

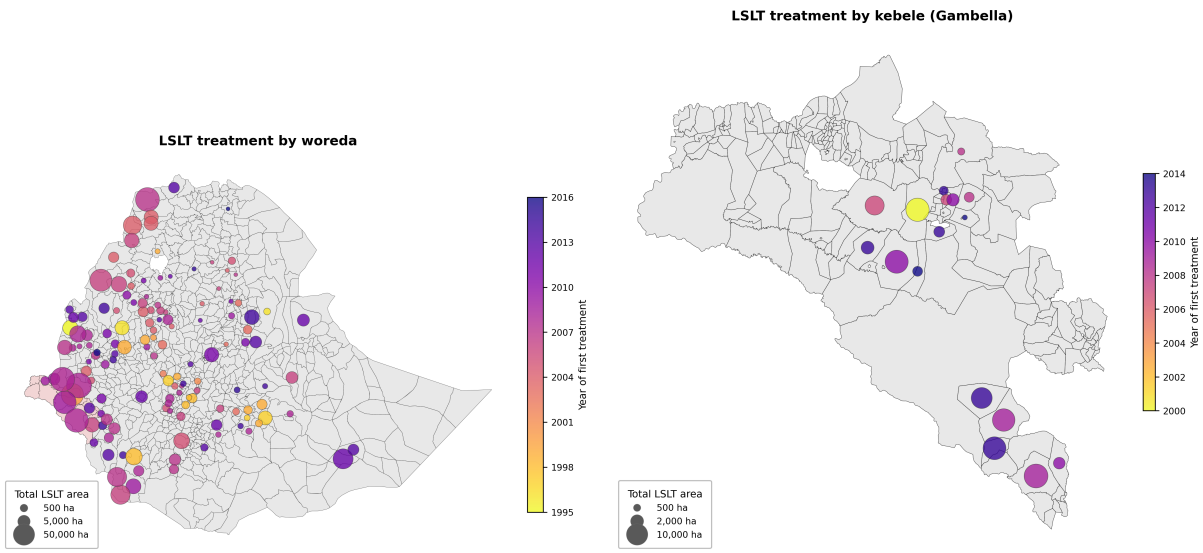
We construct woreda-year and kebele-year panels of environmental outcomes from Google Earth Engine. Pixel-level values are extracted at native sensor resolution (30–500m depending on the product) and aggregated to administrative boundaries as area-weighted means. Cropland NDVI is masked to 2001-baseline cropland pixels (IGBP classes 12, 14) to avoid endogenous land-use change. We also extract annual land cover class shares from the MODIS MCD12Q1 IGBP classification, restricting to forest, savanna, grassland, and cropland.

3.3 TODO Agricultural Sample Survey (AgSS)

Agricultural outcomes come from the Central Statistical Agency's Agricultural Sample Survey (AgSS), a repeated cross-section of field-level agricultural data. We aggregate to woreda-year means for 905 woredas over 2003–2020.

3.4 TODO Gambela Household Survey (LSLT)

We conducted an original household survey covering 2,389 households across 103 kebeles in Gambela region, with two waves (2022–2023). It includes standard consumption and income modules alongside detailed questions on household interactions with LSLTs: employment, land transactions, equipment rental, and perceptions. Treatment exposure is measured as the total LSLT area and count within distance rings (0–5, 5–10, 10–20 km) around each household.



(a) National: Woreda-Level

(b) Gambela: Kebele-Level

Figure 1: Spatial and temporal distribution of LSLT treatment. Circle color indicates year of first treatment (darker = earlier); circle size is proportional to total LSLT area (hectares). Grey polygons are untreated administrative units.

4 Empirical Strategy

We leverage two main identification strategies to estimate the causal effects of LSLTs on local economies and the environment. First, we exploit the plausibly exogenous timing of LSLT entry at the woreda (district) level across Ethiopia in an event study design. We are also able to conduct the event study at the more granular kebele (ward) level in Gambela Region, where we have the exact geolocations of LSLTs. Second, we conduct a cross-sectional analysis using our original survey of 2,389 households in Gambela. Since both waves of these survey are post-treatment, we employ a selection-on-observables design using double-debiased machine learning techniques (Chernozhukov et al., 2018), leveraging the fact the LSLT siting decisions were largely made on observable characteristics such as soil quality and road distance. While this requires stronger identification assumptions, the results are largely consistent with those from the event study analysis, reinforcing the credibility of both

Table 1: Data Sources and Outcome Variables

Variable	Source	Resolution	Period	Unit	Level
Panel A: GEE Remote-Sensing Outcomes					
NPP	MODIS MOD17A3HGF	500m	2001–2023	kgC/m^2/yr	Woreda, Kebele
Cropland NDVI	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda
NDVI (raw)	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda, Kebele
Forest Loss	Hansen GFC v1.12	30m	2001–2023	Fraction	Woreda, Kebele
Tree Cover (VCF)	MODIS MOD44B	250m	2000–2020	% cover	Woreda, Kebele
EVI	MODIS MOD13A1	500m	2001–2023	Index (0–1)	Woreda, Kebele
Burned Area	MODIS MCD64A1	500m	2001–2023	Fraction	Woreda, Kebele
Panel B: Land Cover Shares (Gambella Kebele Level)					
Forest Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Cropland Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Savanna Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Grassland Share	MODIS MCD12Q1	500m	2001–2022	Fraction	Kebele
Panel C: Agricultural Sample Survey (AgSS) Outcomes					
Yield	CSA AgSS	Field survey	2003–2020	kg/ha	Woreda
Cultivated Area	CSA AgSS	Field survey	2003–2020	log ha	Woreda
Fertilizer Use	CSA AgSS	Field survey	2003–2020	Share	Woreda
Improved Seed	CSA AgSS	Field survey	2003–2020	Share	Woreda
Irrigation	CSA AgSS	Field survey	2003–2020	Share	Woreda
Crop Damage	CSA AgSS	Field survey	2003–2020	Share	Woreda
Panel D: Gambella Household Survey (LSIT) Outcomes					
Log Consumption	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Wage Income	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Crop Income	LSIT Survey	Household	2022–2023	Log Birr	HH
Log Harvest (kg)	LSIT Survey	Household	2022–2023	Log kg	HH
Panel E: Treatment Data					
LSLT Parcels	MoA/AG_INV	Polygon	1995–2016	Hectares	Parcel
Woreda Boundaries	CSA 2013	Polygon	—	—	Admin
Kebele Boundaries	CSA 2007	Polygon	—	—	Admin

Notes: GEE outcomes are extracted at native sensor resolution and aggregated to administrative boundaries (woreda or kebele) as area-weighted means. Cropland NDVI is masked to 2001-baseline cropland pixels (IGBP classes 12, 14) to avoid endogenous land-use change. Land cover shares use the IGBP classification from MODIS MCD12Q1. AgSS outcomes are aggregated from field-level survey data to woreda-year means. The LSIT household survey covers Gambella region with two rounds (2022–2023) across 103 kebeles.

sets of findings. Moreover, the survey provides a unique look inside the "black box" of the interactions between households and LSLTs.

4.1 Event Study

Here we leverage the plausibly exogenous timing of LSLT entry into woredas across Ethiopia. While investors have some flexibility in choosing investment locations, the investment is subject to a long series of bureaucratic approvals from the Ethiopia Investment Commission (EIC), the Ministry of Agriculture (MOA) and regional and local authorities. Therefore, conditional on a woreda being targeted for investment, idiosyncratic variation in the time to approval arguably provides suitable identifying variation.

However the staggered, continuous, and repeated nature of our treatment—multiple LSLTs of varying sizes enter woredas over time—presents numerous challenges flagged by the recent literature on negative-weighting bias in two-way fixed effects (TWFE) setups (see Goodman-Bacon, 2021; Roth et al., 2023; Baker et al., 2025). In practice, there are limited robust solutions even for repeated binary treatments, since already treated units cannot be considered "clean controls" with heterogeneous treatment effects. The continuous (in terms of investment size) treatment further complicates matters, requiring stronger assumptions on how units with a higher treatment dose would have evolved had they received a lower dose (Callaway et al., 2024). This restricts the set of estimands for which we can cleanly identify a causal effect.

We therefore present results using different treatment definitions, estimators, and sample restrictions, with careful consideration of what each identifies under which assumptions. Our most cleanly-identified result defines treatment as a dummy variable for the first LSLT entry a woreda receives, using the full sample of woredas. We estimate this with the Callaway and Sant'Anna (2021) (hereafter CS) estimator.[1] This analysis can also be extended to estimate effects of subsequent treatments by restricting controls to untreated units with identical treatment histories following de Chaisemartin and D'Haultfœuille (2024) and Viviano and Bradic (2026).

Moreover, the fact that many approved investments never materialized means that our estimates should be interpreted as intent-to-treat (ITT) estimates.[2] We also control for annual precipitation in our main specifications, although this does not affect the main results substantially.

[1] We prefer CS to the Borusyak et al. (2024) (hereafter BJS) imputation estimator because the latter is highly sensitive to divergent trends between treatment and control from the start of the sample, even when trends have parallelized closer to treatment. By using CS, we are assuming that parallel trends holds for the reference period in question, but allowing for trends to have diverged in the past. We also present results using BJS in the Appendix.

[2] Ongoing analysis is using deep-learning methods applied to Sentinel-2 satellite imagery to identify the establishment of physical LSLTs in treated pixels (following methods developed by Sy & Gehrke, 2026), which would provide valid average treatment on the treated (ATT) estimates if an exogenous component of compliance can be isolated, which we believe to be plausible in this setting.

4.1.1 Gambela Kebele- and Pixel-level event study

We also leverage our more complete administrative data from Gambela—including the precise geolocations of LSLT parcels—to estimate the effects of a more granularly-defined treatment. First, we aggregate to the kebele (ward) as a more granular level of treatment than the woreda: there are 198 kebeles and 10 woredas in Gambela. However, this is only feasible for the remote-sensed outcomes, as we do not have sub-woreda identifiers for the AgSS. Following Kassie et al. (2024), we can also compute pixel-level estimates for remote-sensed outcomes, although we are interested in their effects on households near the concession rather than deforestation on the concession itself. In particular, we extend their analysis by defining a cell as treated if an LSLT has entered within a 5km radius of the cell's boundary and restrict the sample to cells inhabited at baseline (2000). In addition, this analysis helps serve as a validation of the selection on observables approach we employ below.

4.2 Gambela Survey Analysis

Since LSLTs under the government policy stopped in 2016, seven years before data collection, we employ a selection-on-observables design to estimate the causal effect of LSLTs on local economic outcomes. Selection-on-observables designs need to assume that treatment assignment is as good as random conditional on observable covariates. While these assumptions would be difficult to justify ad hoc, investors reportedly based their choice of land almost entirely on soil tests, distance to roads and security. Fortunately, precise geolocated data exists for these variables, allowing us to observe much of the key information upon which they based their locations.

Of course, we don't know how firms used this information to make their location decisions. Since this may have involved a complicated function of a high-dimensional vector of covariates, Double-Debiased Machine Learning (Chernozhukov et al., 2018), hereafter DML, is a natural way to flexibly capture these selection patterns. DML applies orthogonalization and sample-splitting to overcome regularization bias and overfitting, which are common issues in machine learning. As is common, we assume a partially linear model of treatment D_i effects on outcome Y_i. D_i may either be a dummy variable for the allocation of land in a kebele to an LSLT or a continuous measure, such as the share of land in a kebele allocated. We allow outcomes and treatment assignment to depend on (pre-determined) covariates X_i though deterministic nuisance functions g_0 and m_0 respectively.

$$\begin{aligned} Y_i &= \beta D_i + g_0(X_i) + \varepsilon_i \\ D_i &= m_0(X_i) + \nu_i \end{aligned} \tag{1}$$

Under the assumptions that $E[\varepsilon_i|D_i, X_i] = 0$ and $E[\nu_i|X_i] = 0$, the DML procedure provides consistent estimates of the treatment effect, β, using standard machine learning methods to estimate g_0 and m_0. In our case, m_0 is the true relation firms use to decide whether to invest in a site as a function of geophysical and other location-specific factors and g_0 is these factors' direct effects on outcome Y_i. For this procedure to be valid, we assume that we observe the attributes that firms used to choose where to invest and that our estimate $\hat{m}$ well approximates the decision process.

Threats to identification would include missing relevant geophysical attributes, or if LSLTs selected locations based on the receptiveness of local communities or the availability of local labor. We believe that this is unlikely to be the case. First, we have comprehensive soil maps of the area, both from the Ethiopian government and high-resolution gridded models. Second, both government officials and communities were not extensively consulted about LSLTs and given the low engagement of LSLTs with local labor markets, it's unlikely that labor availability played much of a role in siting decisions.

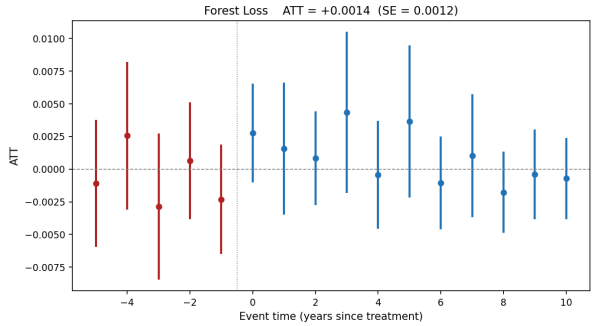
[TO BE CONTINUED]

5 Results

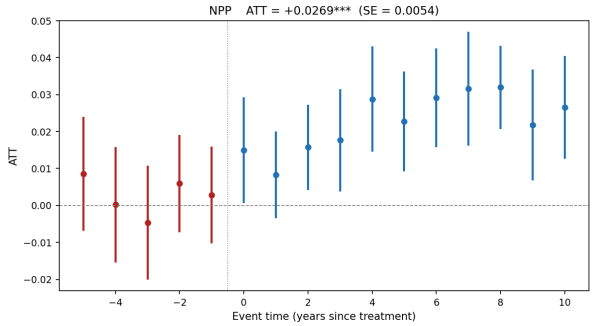
5.1 National Level Event-Study

We estimate the effect of large-scale land transfers on remote-sensing and agricultural outcomes at the woreda level using a staggered difference-in-differences design. We use the Callaway and Sant'Anna (2021) estimator with a varying base period. Our headline event-study figures use the full sample with never-treated woredas as controls; tables additionally report ATT estimates from the ever-treated-only sample (not-yet-treated controls), CS with first-entry area weights, naive TWFE, and the Borusyak et al. (2024) imputation estimator. We drop all woredas with any LSLT parcel pre-2001, as these are never clean controls. Treatment is defined as first LSLT entry with cumulative transfer area ≥ 20 ha. ATT and event-study estimates are restricted to 5 pre-treatment and 10 post-treatment periods. Standard errors are computed via woreda-clustered wild bootstrap (999 iterations).

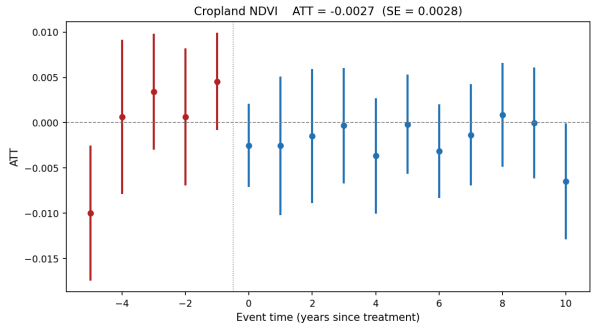
5.1.1 Event Study Plots



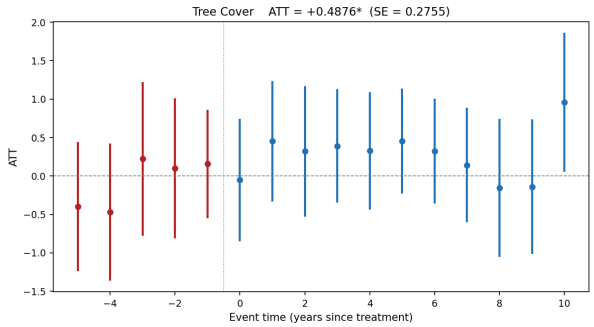
(a) Forest Loss



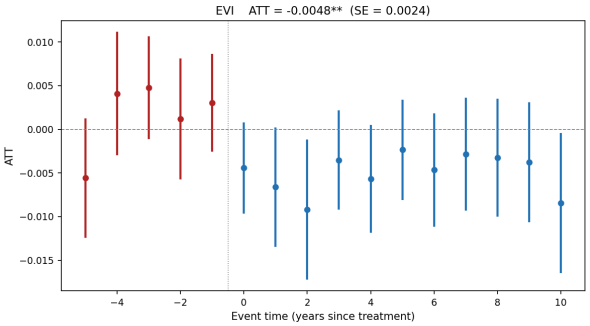
(b) NPP



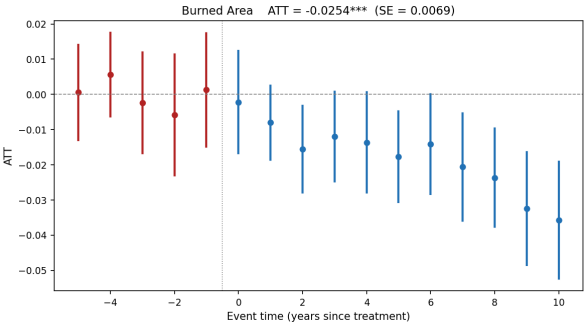
(c) Crop NDVI



(d) Tree Cover (VCF)

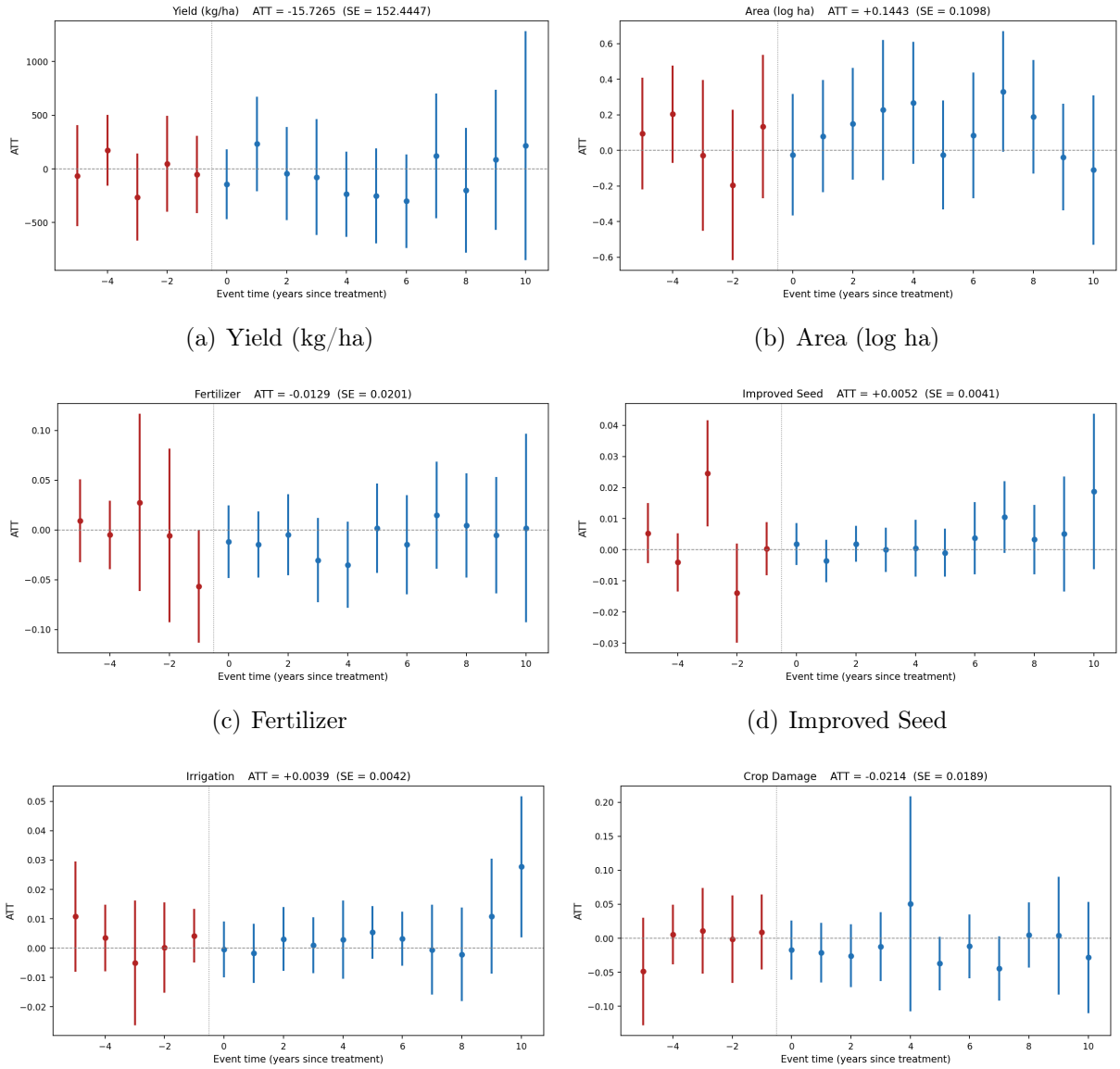


(e) EVI



(f) Burned Area

Figure 2: Callaway-Sant'Anna event-study estimates for GEE remote-sensing outcomes. Full sample including never-treated woredas as controls, varying base period. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



(a) Yield (kg/ha)

(b) Area (log ha)

(c) Fertilizer

(d) Improved Seed

(e) Irrigation

(f) Crop Damage

Figure 3: Callaway-Sant'Anna event-study estimates for AgSS agricultural outcomes. Full sample including never-treated woredas as controls, varying base period, repeated cross-section. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.

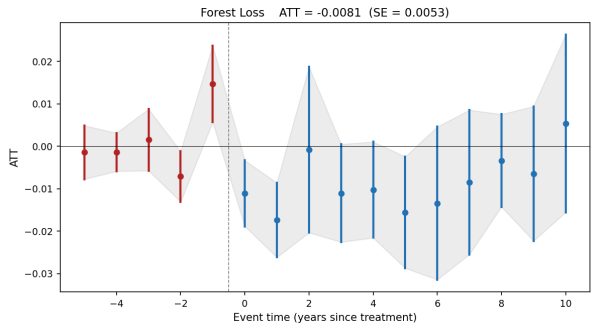
5.1.2 TODO Robustness

We assess sensitivity to sample composition and control group definition across three alternative CS specifications. The ever-treated-only specification restricts the sample to treated woredas with not-yet-treated controls. The PIA-eligible includes woredas with area identified as suitable for LSLTs by the MOA but never treated as controls. The 2nd-entry specification shows results from the *second* treatment of a woreda, using woredas received an initial treatment in the same year but did not (yet) receive a second treatment as controls, following de Chaisemartin and D'Haultfœuille (2024). See Table 10, Table 11, Figure 10 and Table 12 for

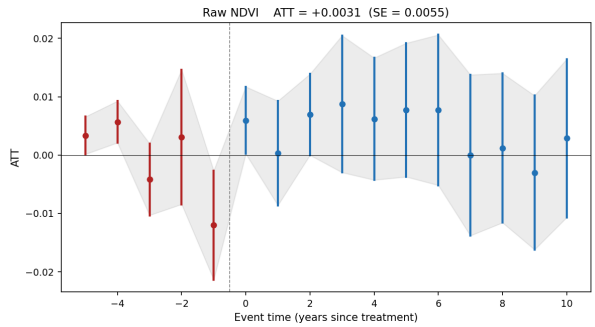
results.

5.2 Kebele-Level Results (Gambela)

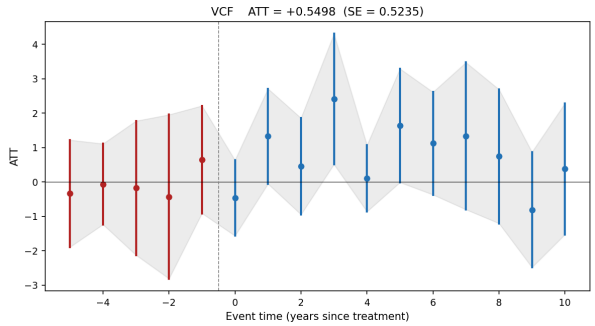
We replicate the event study at the more granular kebele level in Gambela region, where we have geolocated LSLT boundaries matched to individual kebeles. This yields 36 treated kebeles (treatment years 2007–2014) and 162 never-treated kebeles. We estimate CS with both ever-treated-only (not-yet-treated controls) and full-sample (never-treated controls) specifications, clustering standard errors at the woreda level.



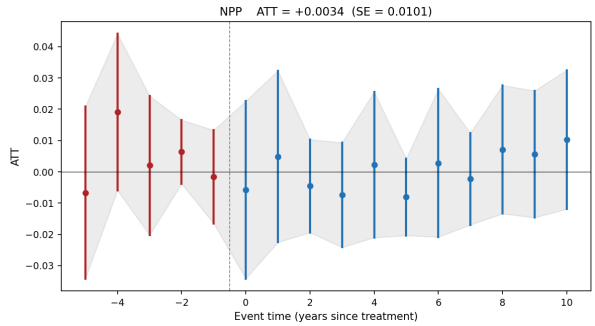
(a) Forest Loss



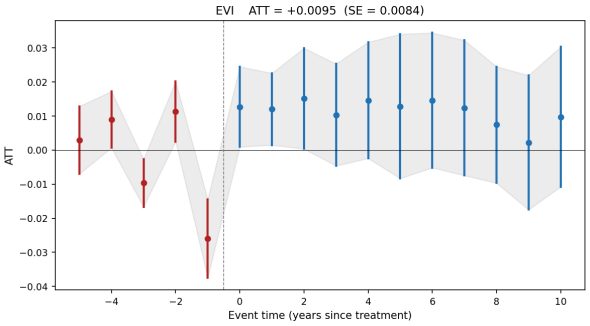
(b) NDVI



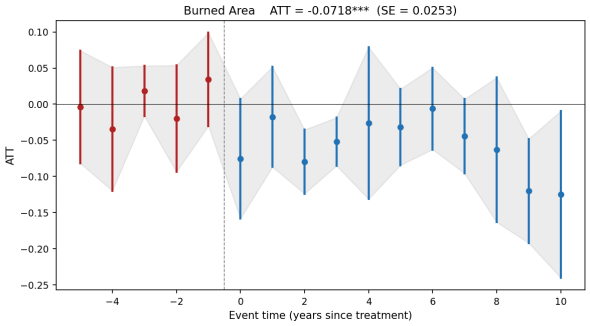
(c) Tree Cover (VCF)



(d) NPP

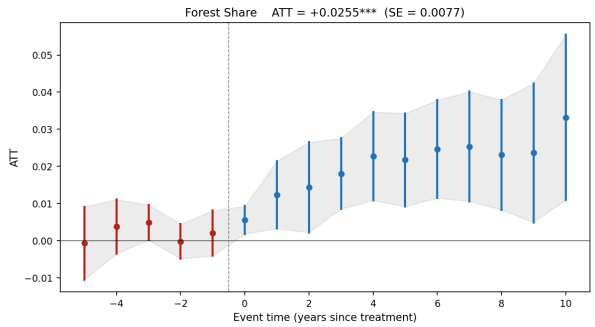


(e) EVI

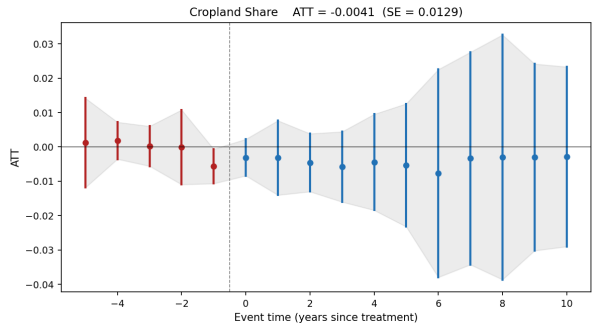


(f) Burned Area

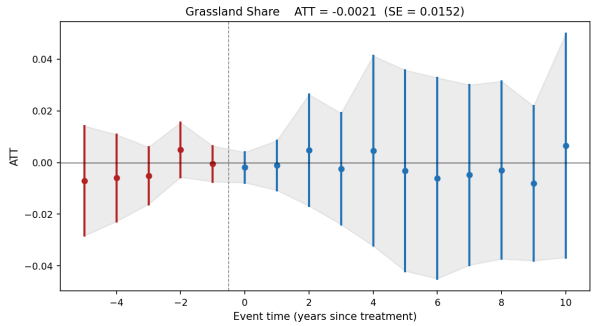
Figure 4: Kebele-level Callaway-Sant'Anna event-study estimates for Gambela region: GEE outcomes. Full sample with never-treated controls; 36 treated kebeles, 162 never-treated. Standard errors clustered at the woreda level.



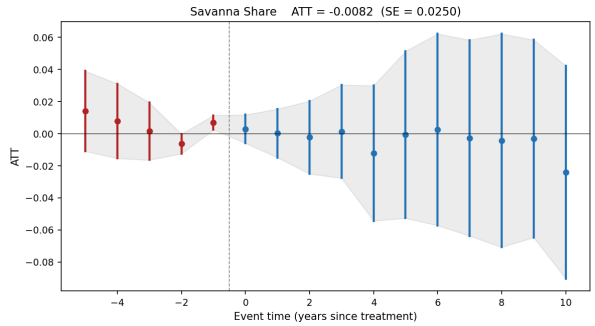
(a) Forest Share



(b) Cropland Share



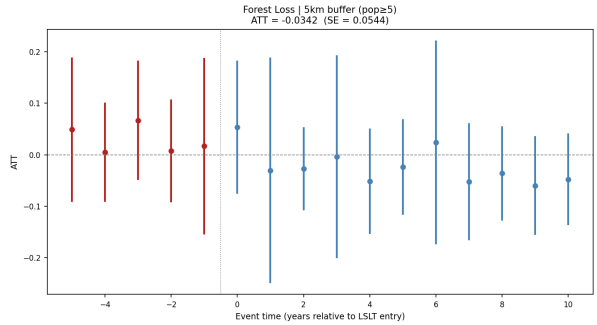
(c) Grassland Share



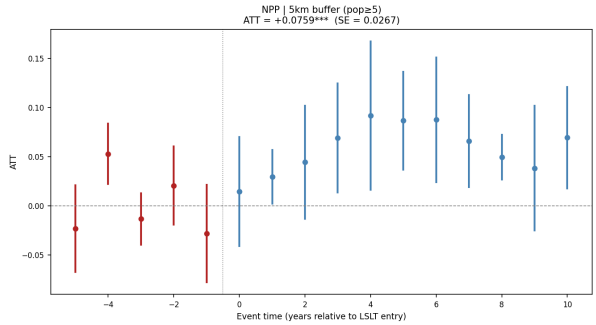
(d) Savanna Share

Figure 5: Kebele-level Callaway-Sant'Anna event-study estimates for Gambela region: land cover shares. Full sample with never-treated controls; 36 treated kebeles, 162 never-treated. Standard errors clustered at the woreda level.

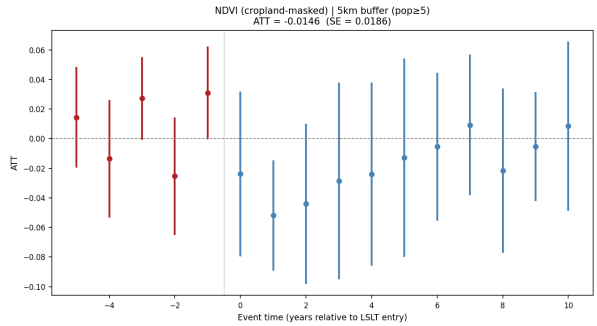
5.3 Pixel-Level Results



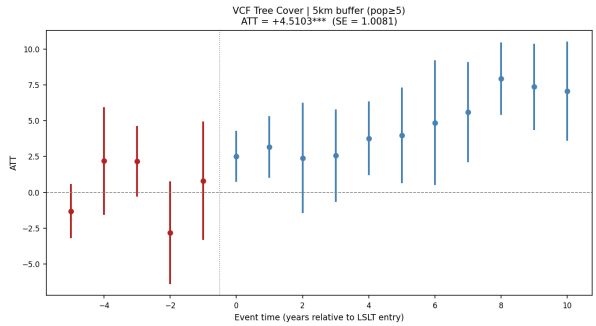
(a) Forest Loss



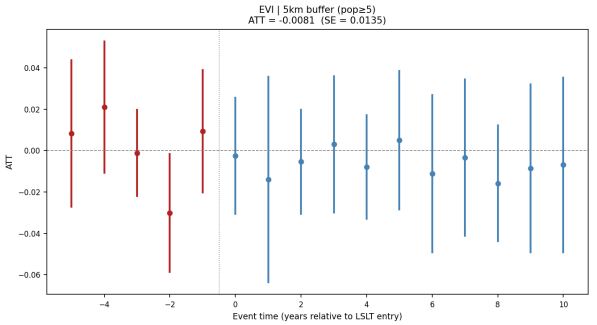
(b) NPP



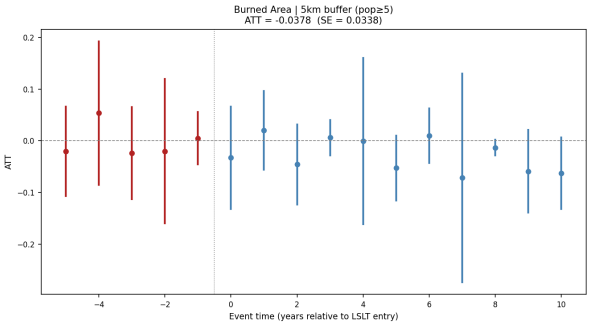
(c) Crop NDVI



(d) Tree Cover (VCF)

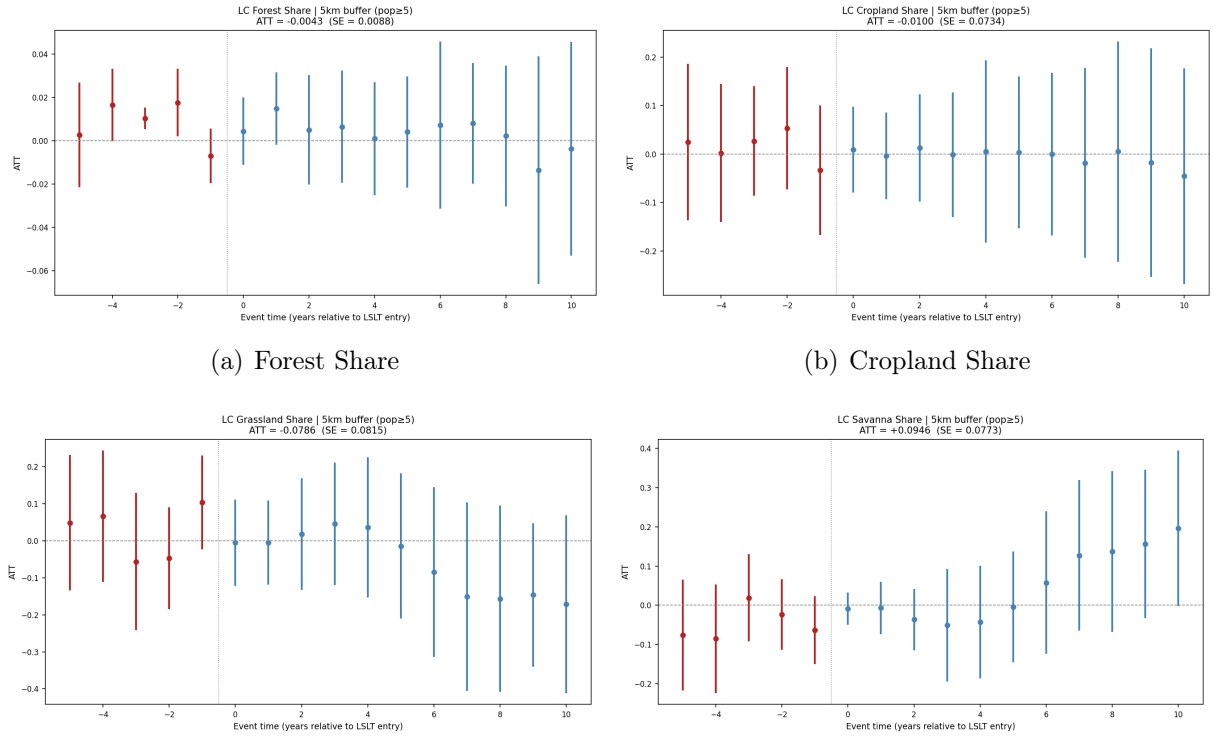


(e) EVI



(f) Burned Area

Figure 6: Pixel-level Callaway-Sant'Anna event-study estimates for GEE remote-sensing outcomes. Treatment defined as LSLT entry within 5 km of pixel centroid, restricted to pixels inhabited at baseline (2000). Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



(a) Forest Share

(b) Cropland Share

(c) Grassland Share

(d) Savanna Share

Figure 7: Pixel-level Callaway-Sant'Anna event-study estimates for GEE land classification outcomes. Treatment defined as LSLT entry within 5 km of pixel centroid, restricted to pixels inhabited at baseline (2000). Whiskers show 95% CIs; overall ATT and SE reported in each panel title.

5.4 Gambela Survey

We estimate the causal effects of LSLT proximity on household welfare using Double-Debiased Machine Learning (DML) with a partially linear regression model. Treatment is measured as the total LSLT area (in 1,000 ha) within three distance rings around each household: 0–5 km, 5–10 km, and 10–20 km. A stacking ensemble learner (XGBoost + Random Forest + Elastic Net → Ridge meta-learner) is used for nuisance estimation; standard errors are clustered at the kebele level ($n = 103$ clusters). Results are stable across alternative nuisance learners (see robustness tables in the appendix).

5.4.1 Spatial Gradient in Labor Market Effects

The headline finding is a pronounced spatial gradient in labor market integration. Table 2 reports estimates for wage income and employment outcomes. Households located 5–10 km from LSLT sites experience a significant increase in log wage income ($\hat{\theta} = 0.137$, $p < 0.05$) and in the probability of having any wage income ($\hat{\theta} = 0.009$, $p < 0.10$). By contrast, households within 0–5 km of an LSLT show no wage income gain and a directionally negative consumption effect The far ring (10–20 km) shows uniformly null results. This pattern—negative or null effects at close range, positive labor market effects at intermediate distances,

and no effects far away—is consistent with a demand-side labor market channel that reaches intermediate communities while the most exposed households bear resource and displacement costs.

Figure 8 and Figure 9 shows descriptive evidence on household interactions with LSLTs by distance, corroborating the spatial gradient: engagement through wage labor is most common at intermediate distances, while direct land-use conflicts are concentrated among the nearest households.

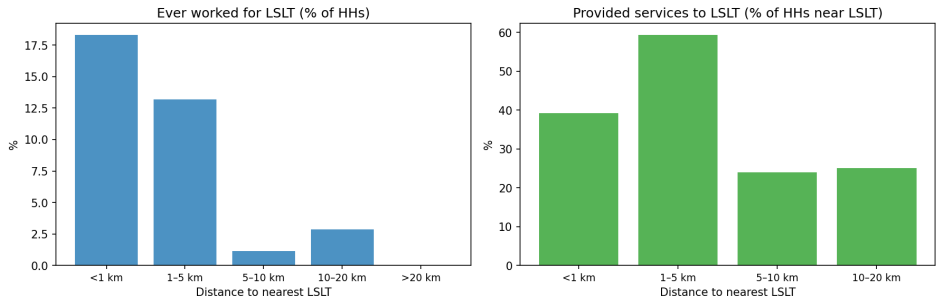


Figure 8: Household interactions with LSLT sites by distance ring. Shares are conditional on being within 20 km of at least one LSLT parcel. Source: Gambela household survey.

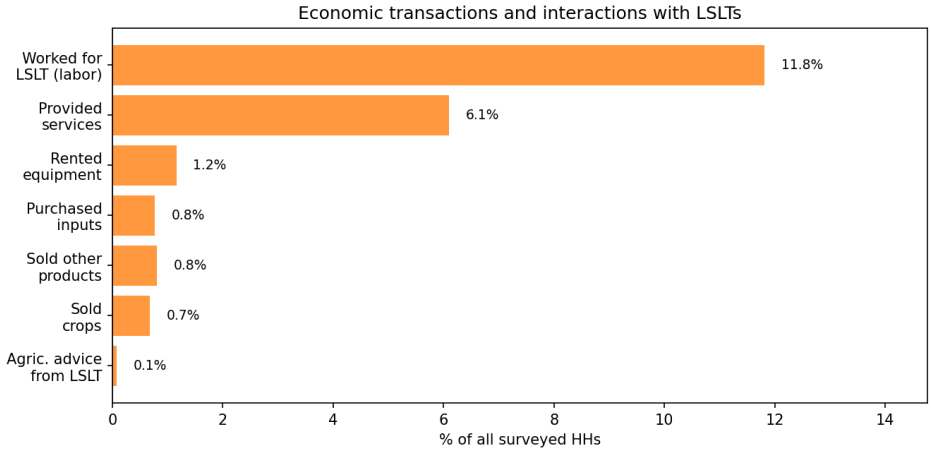


Figure 9: Household interactions with LSLT sites by type. Source: Gambela household survey.

5.4.2 Main DML Results Tables

5.4.3 Robustness

Table 13–Table 16 replicate the key outcomes under alternative nuisance learners (XGBoost, Random Forest, Elastic Net) and post-double Lasso OLS. The 5–10 km wage income effect is robust across all specifications. The directionally negative consumption effect at 0–5 km reaches conventional significance only in the post-lasso benchmark, reflecting lower power from kebele-clustered variance estimation with 103 clusters under the DML specs.

Table 2: DML Estimates: Wages and Employment

	Log wage income	Has wage income	Log business income
0–5 km	+0.010	−0.009	+0.196
	(0.128)	(0.014)	(0.124)
5–10 km	+0.137**	+0.009*	+0.022
	(0.061)	(0.005)	(0.060)
10–20 km	+0.005	+0.000	+0.002
	(0.019)	(0.002)	(0.013)
N	4,814	4,814	4,814

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Stacking ensemble (XGBoost + RF + Elastic Net → Ridge meta-learner, $n_{\text{rep}} = 5$, $k = 3$ folds).

Table 3: DML Estimates: Agricultural Outcomes

	Log crop income	Log harvest (kg)	# crops
0–5 km	+0.078	−0.128	+0.003
	(0.166)	(0.115)	(0.057)
5–10 km	+0.083	+0.041	+0.048
	(0.058)	(0.046)	(0.036)
10–20 km	−0.007	−0.008	+0.004
	(0.020)	(0.012)	(0.008)
N	4,814	4,814	3,819

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Stacking ensemble (XGBoost + RF + Elastic Net → Ridge meta-learner, $n_{\text{rep}} = 5$, $k = 3$ folds).

Table 4: DML Estimates: Consumption and Total Income

	Log consumption	Log total HH income
0–5 km	−0.032	+0.072
	(0.026)	(0.111)
5–10 km	−0.013	+0.086
	(0.010)	(0.064)
10–20 km	+0.004	−0.025
	(0.004)	(0.019)
N	4,814	4,814

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Stacking ensemble (XGBoost + RF + Elastic Net → Ridge meta-learner, $n_{\text{rep}} = 5$, $k = 3$ folds).

Table 5: DML Estimates: Environmental Income

	Log env income	Has env income
0–5 km	+0.090	+0.009
	(0.084)	(0.010)
5–10 km	−0.003	−0.004
	(0.037)	(0.005)
10–20 km	+0.009	+0.000
	(0.015)	(0.001)
N	4,814	4,814

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Stacking ensemble (XGBoost + RF + Elastic Net → Ridge meta-learner, $n_{\text{rep}} = 5$, $k = 3$ folds).

6 TODO Conclusion

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7 Appendix

7.1 Additional Tables and Figures

Table 6: Summary of quantitative LSLT impact studies

Paper	Context	Method	Key finding
Economic outcomes			
Ali et al. (2019)	Ethiopia	PSM, cross-section	No job creation; modest tech adoption
Shete and Rutten (2015)	Ethiopia (Oromia)	PSM, cross-section	Negative food security, income
Deininger and Xia (2016)	Mozambique	PSM	Positive inputs; negative well-being
Bottazzi et al. (2018)	Sierra Leone	PSM	Higher income; heterog. by gender
Anti (2021)	Cambodia	TWFE	Negative consumption; wage shift
Jiao et al. (2015)	Cambodia	Matching	Negative income, land, livestock
Bunte et al. (2018)	Liberia	TWFE	No growth effects
Müller et al. (2021)	Multi-country	Meta-analysis	Increased production, food insecurity
Akalu et al. (2025)	Multi-country	Meta-analysis (143 studies)	Modest positive overall; heterog. by country
Sy and Gehrke (2026)	Senegal	TWFE	Increased consumption and wage labor
Environmental outcomes			
Davis et al. (2015)	Cambodia	Matching	Substantial forest loss
Davis et al. (2020)	15 countries	Matching	Forest loss; heterogeneous
Rulli et al. (2019)	Indonesia	Matching	Forest loss, fragmentation
Liao et al. (2020)	Eth., Cambodia, Peru, Liberia	Simulation	Projected large forest loss
Social outcomes			
Wegenast et al. (2022)	Multi-country	Survey	Reduced trust
Khadjavi et al. (2021)	Multi-country	Lab-in-field	Increased trust/social capital
Hajjar et al. (2020)	Ethiopia	Qualitative (4 farms)	Land use change, labor reallocation

Table 7: Summary Statistics — National Woreda Panel

	N	Mean	SD	Min	Max
Panel A: GEE Remote Sensing Outcomes					
NPP (kgC/m^2/yr)	15,755	0.834	0.421	0.000	2.133
Crop NDVI	14,191	0.752	0.073	0.233	0.922
Raw NDVI	15,847	0.695	0.160	0.090	0.922
Forest Loss (fraction)	15,847	0.009	0.019	0.000	0.369
Tree Cover (VCF %)	15,847	13.592	11.858	0.000	77.739
EVI	15,847	0.477	0.134	0.048	0.715
Burned Area (fraction)	15,847	0.036	0.103	0.000	0.913
Panel B: AgSS Agricultural Outcomes					
Yield (kg/ha)	8,044	2,306	2,011	0.000	9,983
Fertilizer Use (share)	9,212	0.363	0.210	0.000	0.907
Improved Seed (share)	9,212	0.027	0.044	0.000	0.542
Irrigation (share)	9,212	0.023	0.059	0.000	0.762
Crop Damage (share)	9,022	0.434	0.173	0.000	5.160
Mean Plot Area (ha)	8,929	0.109	0.082	0.000	2.323

GEE outcomes cover 689 woredas, 2001–2023. AgSS outcomes cover 905 woredas, 2003–2020.

Table 8: Summary Statistics — Gambella Kebele Panel

	N	Mean	SD	Min	Max
Panel A: GEE Remote Sensing Outcomes					
NPP (kgC/m^2/yr)	4,554	0.612	0.399	0.000	1.777
Crop NDVI	3,128	0.808	0.036	0.589	0.906
Raw NDVI	4,554	0.824	0.048	0.593	0.936
Forest Loss (fraction)	4,554	0.011	0.029	0.000	0.607
Tree Cover (VCF %)	4,554	25.321	16.517	2.837	79.057
EVI	4,554	0.608	0.063	0.363	0.782
Burned Area (fraction)	4,554	0.230	0.245	0.000	1.000
Panel B: Land Cover Shares					
Forest Share	4,554	0.171	0.325	0.000	1.000
Cropland Share	4,554	0.028	0.062	0.000	0.633
Savanna Share	4,554	0.502	0.356	0.000	1.000
Grassland Share	4,554	0.264	0.289	0.000	1.000

198 kebeles in Gambella region, 2001–2023. Land cover shares from MODIS MCD12Q1.

Table 9: Summary Statistics — Gambella Household Survey

	N	Mean	SD	Min	Max
Panel A: Key Outcomes					
Log Consumption	4,833	7.754	0.690	1.204	11.578
Log Total HH Income	4,833	9.035	4.155	0.000	15.646
Log Wage Income	4,833	2.722	4.753	0.000	13.541
Log Crop Income	4,833	3.594	4.874	-12.172	15.201
Log Harvest (kg)	4,833	5.658	3.115	0.000	14.457
Log Crop Revenue	4,833	4.008	4.807	0.000	15.202
Panel B: LSLT Exposure Measures					
LSLT Area within 5 km (ha)	4,833	0.168	0.195	0.000	0.810
LSLT Count within 5 km	4,833	5.042	6.399	0.000	32.000
LSLT Area within 10 km (ha)	4,833	0.219	0.208	0.000	0.788
LSLT Count within 10 km	4,833	16.246	18.828	0.000	79.000
Min. Distance to LSLT (km)	4,814	0.411	0.537	0.000	3.645
Investor Share of Kebele Area	4,833	0.217	0.268	0.000	0.883

2,527 households across 2 survey rounds. Distance converted from decimal degrees ($1^\circ \approx$ 111 km).

Table 10: Effect of LSLT Entry on Remote-Sensing Outcomes

	CS	CS (weighted)	TWFE	BJS
NPP	$+0.0269^{***}$	$+0.0289^{***}$	$+0.0304^{***}$	$+0.0084$
	(0.0054)	(0.0085)	(0.0054)	(0.0061)
Cropland NDVI	-0.0027	-0.0005	$+0.0016$	$+0.0028$
	(0.0028)	(0.0073)	(0.0016)	(0.0019)
NDVI (raw)	-0.0046^{*}		-0.0005	-0.0023
	(0.0026)		(0.0015)	(0.0018)
Forest Loss	$+0.0014$	$+0.0017$	-0.0004	$+0.0057^{***}$
	(0.0012)	(0.0030)	(0.0009)	(0.0016)
Tree Cover (VCF)	$+0.4876^{*}$	$+0.4714$	$+0.2593$	-0.0769
	(0.2755)	(0.6766)	(0.2046)	(0.2771)
EVI	-0.0048^{**}		$+0.0003$	-0.0042^{**}
	(0.0024)		(0.0016)	(0.0019)
Burned Area	-0.0254^{***}		-0.0180^{***}	-0.0126^{***}
	(0.0069)		(0.0038)	(0.0040)
Joint pre-trend p	0.048–0.920†	0.006–0.672†	—	0.067–0.859

Notes: Each cell reports the overall ATT (standard error) for the indicated outcome–estimator pair. CS: Callaway-Sant'Anna with varying base period, not-yet-treated and never-treated controls, full sample ($N = 689$ woredas); CHIRPS precipitation as covariate. CS (weighted): CS with first-entry area weights (R `did` package). TWFE: two-way fixed effects (entity + year), full sample. BJS: Borusyak-Jaravel-Spiess imputation estimator, ever-treated only. Standard errors clustered by woreda. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 11: Effect of LSLT Entry on Agricultural Outcomes (AgSS)

	CS	CS (weighted)	TWFE	BJS
Yield (kg/ha)	-15.7	-69.2	$+388.4^{***}$	$+193.1$
	(152.4)	(269.6)	(115.9)	(146.6)
Area (log ha)	$+0.144$	-0.325		$+0.196^{**}$
	(0.110)	(0.258)		(0.074)
Fertilizer	-0.0129	$+0.0434$	-0.0072	$+0.0035$
	(0.0201)	(0.0422)	(0.0081)	(0.0093)
Improved Seed	$+0.0052$	$+0.0006$	$+0.0040$	$+0.0046^{*}$
	(0.0041)	(0.0037)	(0.0028)	(0.0024)
Irrigation	$+0.0039$	$+0.0197^{*}$	$+0.0062^{**}$	-0.0010
	(0.0042)	(0.0106)	(0.0029)	(0.0035)
Crop Damage	-0.0214	-0.0099	$+0.0089$	$+0.0077$
	(0.0189)	(0.0294)	(0.0129)	(0.0143)

Notes: Same estimator specifications as Table 10. Outcomes from the Agricultural Sample Survey (AgSS) repeated cross-section at the woreda level. Standard errors clustered by woreda. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 12: Robustness: CS Under Alternative Samples and Controls (GEE Outcomes)

	CS ever-treated	CS PIA controls	CS 2nd-entry
NPP	+0.0096	−0.0056	+0.0141
	(0.0069)	(0.0102)	(0.0325)
Cropland NDVI	+0.0000	−0.0028	−0.0031
	(0.0035)	(0.0035)	(0.0082)
NDVI (raw)	−0.0056	−0.0034	+0.0125
	(0.0040)	(0.0029)	(0.0083)
Forest Loss	+0.0021	+0.0044**	−0.0067**
	(0.0021)	(0.0020)	(0.0030)
Tree Cover (VCF)	+0.5026	+0.5007	−5.6710***
	(0.3200)	(0.3430)	(1.6798)
EVI	−0.0089**	−0.0077**	+0.0131
	(0.0036)	(0.0036)	(0.0102)
Burned Area	−0.0125	−0.0092	−0.0059
	(0.0078)	(0.0102)	(0.0173)
Joint pre-trend p (Wald χ^2)			
NPP	0.659	0.052	0.687
Cropland NDVI	0.051	0.630	0.025†
NDVI (raw)	0.712	0.712	0.180
Forest Loss	0.865	0.327	0.086
Tree Cover (VCF)	0.898	0.628	0.772
EVI	0.498	0.591	0.106
Burned Area	0.995	0.693	0.738

Notes: Callaway-Sant'Anna estimator with varying base period. Full sample: all woredas including never-treated as controls. PIA controls: ever-treated plus never-treated woredas with positive Potential Investment Area. 2nd-entry: treatment redefined as 2nd cumulative threshold crossing; single-entry woredas as controls. CHIRPS precipitation as covariate. Standard errors clustered by woreda. †: joint pre-trend $p < 0.05$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

(a) CS ever-treated only

(b) PIA-eligible controls

(c) 2nd-entry (treatment history)

(d) BJS imputation

Figure 10: Forest Loss: Robustness across alternative specifications. Panel (a) restricts to ever-treated woredas with not-yet-treated controls. Panel (b) restricts never-treated controls to woredas with positive Potential Investment Area. Panel (c) redefines treatment as the second entry, conditioning on prior treatment history. Panel (d) uses the BJS imputation estimator for comparison.

Table 13: Robustness: XGBoost

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH income
0–5 km	−0.059	+0.001	−0.009	+0.026	−0.011	+0.043
	(0.200)	(0.013)	(0.182)	(0.115)	(0.025)	(0.156)
5–10 km	+0.088	+0.015**	+0.139**	+0.107*	−0.008	+0.095
	(0.073)	(0.007)	(0.069)	(0.059)	(0.011)	(0.071)
10–20 km	+0.004	−0.002	−0.038	−0.033	+0.004	−0.020
	(0.033)	(0.003)	(0.033)	(0.024)	(0.005)	(0.025)
N	4,814	4,814	4,814	4,814	4,814	4,814

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: XGBoost ($n_{\text{rep}} = 5$, $k = 3$ folds).

Table 14: Robustness: Random Forest

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH income
0–5 km	−0.076	−0.006	+0.027	+0.003	−0.027	−0.115
	(0.195)	(0.017)	(0.161)	(0.134)	(0.026)	(0.182)
5–10 km	+0.095	+0.008	+0.088	+0.070	−0.005	+0.095
	(0.072)	(0.006)	(0.066)	(0.050)	(0.013)	(0.066)
10–20 km	+0.011	−0.000	−0.035	−0.026	+0.003	-0.057^{**}
	(0.032)	(0.003)	(0.025)	(0.016)	(0.004)	(0.026)
N	4,814	4,814	4,814	4,814	4,814	4,814

Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Random Forest ($n_{\text{rep}} = 5$, $k = 3$ folds).

Table 15: Robustness: Elastic Net

	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH income
0–5 km	+0.039	+0.012	+0.075	−0.155	−0.007	+0.042
	(0.146)	(0.011)	(0.168)	(0.123)	(0.020)	(0.112)
5–10 km	$+0.144^{***}$	$+0.014^{***}$	+0.017	−0.013	−0.005	+0.038
	(0.046)	(0.005)	(0.056)	(0.038)	(0.011)	(0.053)
10–20 km	+0.013	+0.002	−0.015	−0.012	+0.002	+0.003
	(0.018)	(0.002)	(0.013)	(0.009)	(0.002)	(0.013)
N	4,814	4,814	4,814	4,814	4,814	4,814

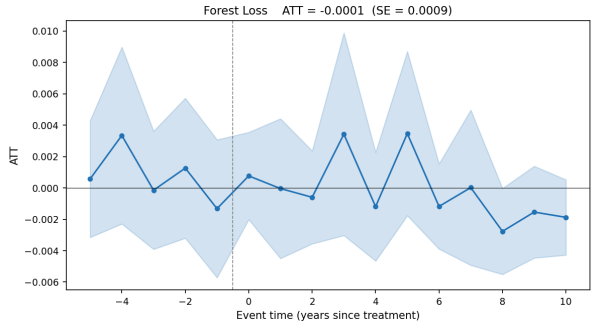
Notes: Coefficients from Double ML–PLR with kebele-clustered standard errors in parentheses. Treatment variables measure LSLT area within each distance ring (1,000 ha). Controls: soil properties, terrain, climate, vegetation, road access, potential investment area, safety index, ethnicity indicators, and survey wave. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Nuisance learner: Elastic Net with CV ($n_{\text{rep}} = 5$, $k = 3$ folds).

Table 16: Robustness: Post-Double Lasso OLS

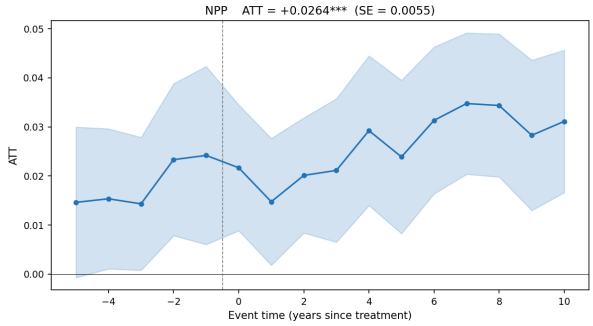
	Log wage income	Has wage	Log crop income	Log harvest	Log cons.	Log HH income
0–5 km	−0.342	−0.033	+0.058	−0.156	−0.057**	−0.079
	(0.219)	(0.020)	(0.157)	(0.114)	(0.025)	(0.141)
5–10 km	+0.168**	+0.016**	+0.079	+0.077	+0.009	−0.022
	(0.082)	(0.008)	(0.077)	(0.056)	(0.012)	(0.072)
10–20 km	+0.001	+0.000	−0.005	−0.012	+0.005	+0.027
	(0.024)	(0.002)	(0.026)	(0.019)	(0.003)	(0.026)
N	4,814	4,814	4,814	4,814	4,814	4,814

Notes: Coefficients from Double Post-Lasso OLS with kebele-clustered standard errors in parentheses. Controls selected via LassoCV from the union of variables selected in the outcome and each treatment equation. Treatment variables measure LSLT area within each distance ring (1,000 ha). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

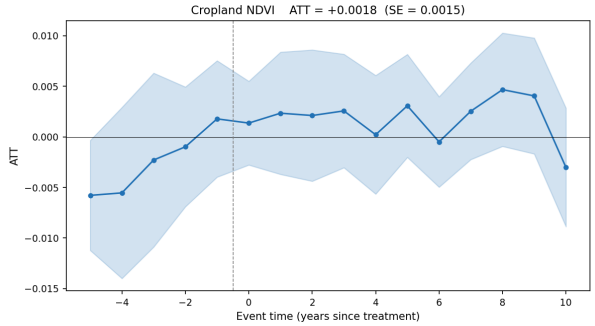
7.2 BJS Imputation Event-Study Estimates



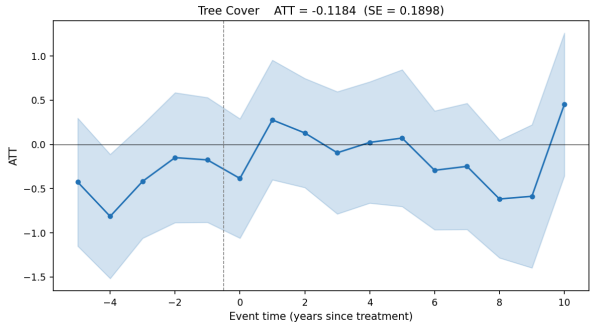
(a) Forest Loss



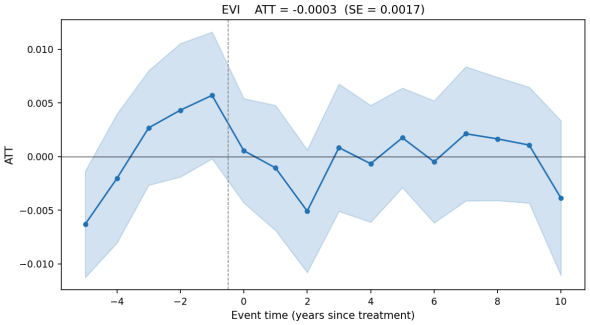
(b) NPP



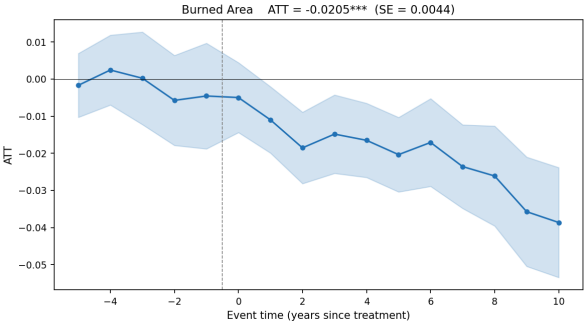
(c) Crop NDVI



(d) Tree Cover (VCF)

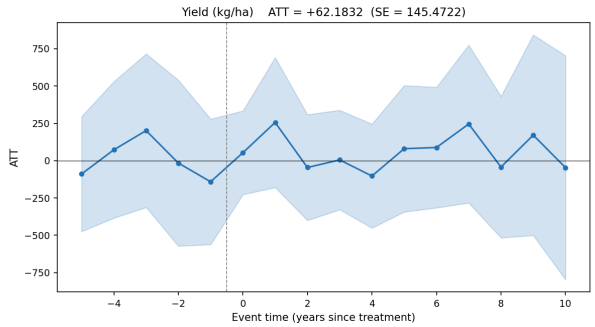


(e) EVI

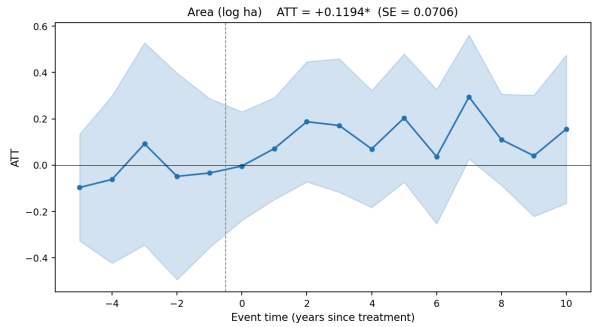


(f) Burned Area

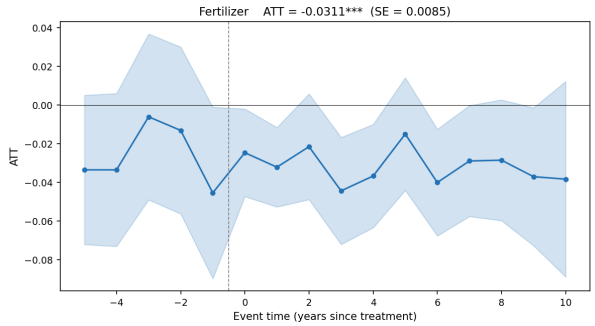
Figure 11: BJS imputation event-study estimates for GEE remote-sensing outcomes. Full sample including never-treated woredas as controls. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.



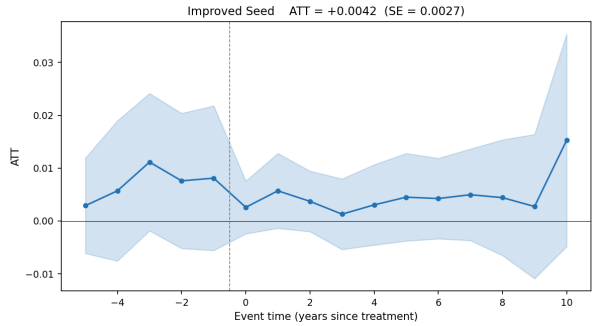
(a) Yield (kg/ha)



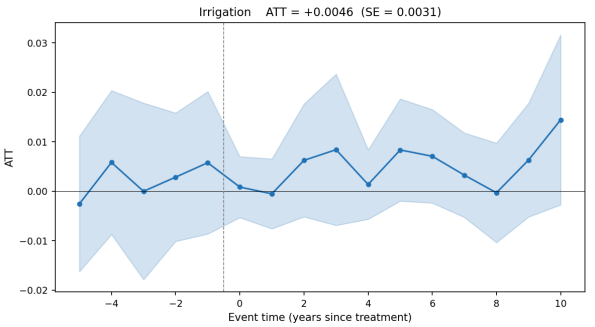
(b) Area (log ha)



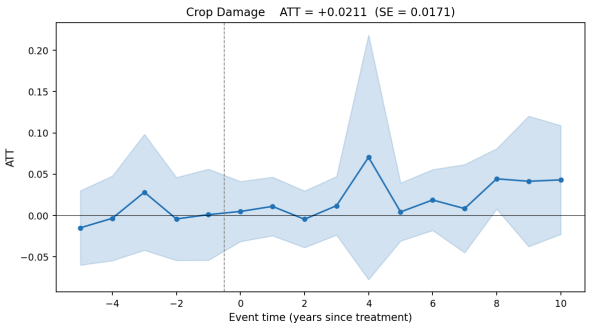
(c) Fertilizer



(d) Improved Seed



(e) Irrigation



(f) Crop Damage

Figure 12: BJS imputation event-study estimates for AgSS agricultural outcomes. Full sample including never-treated woredas as controls. Whiskers show 95% CIs; overall ATT and SE reported in each panel title.